RESEARCH PAPER / INTERNAL KNOWLEDGE BASE SEARCH

PHASE 2 — DATABASE FUNCTIONALITY AND AUTOMATION

Phase 2 extends the research-paper database with reusable stored routines, automated record maintenance, and an audit trail. These additions move frequently used operations into PostgreSQL, making them callable in a consistent manner and recording data changes at database level.

Implemented Functionalities

Component	Implementation	Outcome
Paper insertion procedure	`add_paper` encapsulates insertion of a paper record as a stored procedure.	Provides a reusable database entry point for paper creation.
Paper deletion procedure	`delete_paper` encapsulates deletion by paper identifier.	Provides a consistent database operation for removing a paper.
Chunk-count function	`count_paper_chunks` returns the number of content chunks associated with a paper.	Enables direct retrieval of per-paper chunk statistics.
Citation-count function	`count_paper_citations` returns the number of citation records associated with a paper.	Enables direct retrieval of per-paper citation statistics.
Automatic timestamp update	A `BEFORE UPDATE` trigger invokes timestamp-maintenance logic for paper records.	Maintains modification timestamps without requiring each caller to set them manually.
Change-audit trigger	An `AFTER INSERT / UPDATE / DELETE` trigger records paper changes in the audit table.	Maintains an audit history of paper data changes.

Verification Results

The database routines were exercised against the project database. The chunk-count and citation-count functions returned results for a sample paper. A temporary paper record was inserted through `add_paper`, and the audit trigger recorded INSERT, UPDATE, and DELETE events. The temporary record was then removed using `delete_paper`, while the audit history remained available.

Test	Observed result
Chunk and citation counts	Functions executed and returned counts for a sample paper.
Paper lifecycle	Temporary paper inserted, updated, and deleted during testing.
Audit history	Audit entries were created for INSERT, UPDATE, and DELETE operations.

Technical Contribution

The stored procedures and functions provide reusable database operations, while triggers enforce automatic behavior at the point of data modification. The audit mechanism separates the current paper state from the history of changes, allowing operations to be inspected after the affected record has been changed or removed.

The database also contains pgvector-based vector storage and cosine-distance search infrastructure. Meaningful semantic retrieval depends on storing embeddings generated from the content with a model whose output dimension matches the database vector column.

Conclusion

The Phase 2 additions demonstrate stored procedures, user-defined functions, trigger-driven timestamp maintenance, and audit logging in PostgreSQL. Together, these features improve operational consistency and traceability while providing a database foundation for research-paper retrieval.